# Parametric & Sequences Test, Day 1

Cram sheet

**Wednesday, no calculator:** 5 parametric curve questions + 1 sequence question (first five terms or recursive). 16 points.

## How far to simplify (every rectangular answer)

- **Whole graph shows** (circle, ellipse, both branches): done once $t$ is gone, e.g. $\frac{x^2}{4} - y^2 = 1$. Solving for $y$ is also right, only with $\pm$.
- **Part of it, a domain can cut it** (piece of a parabola, one left-right branch): add "for $x$ in $[a,b]$", e.g. $\frac{x^2}{4} - y^2 = 1$ for $x$ in $[2,\infty)$.
- **Part of it, a domain can't** (top or bottom of an up-down hyperbola): solve for $y$, keep one sign, e.g. $y = 2\sqrt{\frac{x^2}{9} + 1}$.
- **Small piece of a branch:** solve for $y$ *and* add "for $x$ in". Never "for $y$ in" or "$y \geq$".
- Question says **SIMPLIFIED**: all the way, $y$ alone ($y = -\frac{1}{3}x^2$). No "simplify": any form with $t$ gone; don't multiply out.

## Parametric curves

1. **$t$-chart first.** Brackets: plug in both ends, solid dots. Parentheses: values inside.
2. Solve for $t$ where it gives **one answer** (no $\pm$); substitute into the other.
3. Domain: smallest and largest $x$ in the chart, "for $x$ in $[a,b]$". Watch turning points ($x = t^2 \geq 0$, $x = \sqrt{t} \geq 0$).
4. Any form is fine once $t$ is gone; no multiplying out. **SIMPLIFIED** means $y$ alone. No domain written means all real $x$.
5. Direction arrows **only when the question asks**: lowest $t$ to highest. End arrowheads only when the curve keeps going. Asymptotes optional, dashed if drawn.

| **What the $t$-chart shows** | **Final form** (whole graph: done; domain can cut it: "for $x$ in"; can't: solve for $y$, one sign) |
|---|---|
| full circle / ellipse | $\frac{x^2}{9} + \frac{y^2}{25} = 1$, no domain |
| one hyperbola branch (restricted $t$) | solve for $y$, keep that branch's sign ($+\sqrt{}$ top, $-\sqrt{}$ bottom): $y = 2\sqrt{\frac{x^2}{9} + 1}$; never "$y \geq$"; any equivalent form |
| both branches (no restriction) | $\frac{x^2}{4} - \frac{y^2}{9} = 1$, done (or $y = \pm 3\sqrt{\frac{x^2}{4} - 1}$, never $+\sqrt{}$ alone) |
| one left-right branch | $\frac{x^2}{4} - y^2 = 1$ for $x$ in $[2,\infty)$; never "for $y$ in" |
| reciprocal $x = t+2,\ y = \frac{1}{t+2}$ | $y = \frac{1}{x}$; "for all $x \neq 0$" is good practice (it comes with the equation) |

## Two parametric representations

Give both $x =$ and $y =$ ($t = 4 - x$ is not parametric form). $y = (x-2)^2 + 1$: $x = t,\ y = (t-2)^2 + 1$ and $x = t+2,\ y = t^2 + 1$, for $t$ in $(-\infty,\infty)$ (or the restricted interval). Not in vertex form? Pick any shift $x = t + h$, substitute, simplify.

## Identities for eliminating $t$

$\sin^2 t + \cos^2 t = 1$, $1 + \tan^2 t = \sec^2 t$, $1 + \cot^2 t = \csc^2 t$.

## The sequence question: first five terms or recursive

- **First five terms:** plug in $n = 1,2,3,4,5$. Answer is a list with commas, reduced, no ellipsis. $(-1)^n$: odd $n$ negative.
- **Recursive:** the formula starts where it says ($n \geq 2$, or $n \geq 3$ when two terms are given); $a_{n-1}$ is the previous term, $a_{n-2}$ the one before it.

Ex: $a_1 = -1,\ a_2 = 3,\ a_n = a_{n-1}a_{n-2} \to -1, 3, -3, -9, 27$ $\quad$ $\{(-1)^n \frac{n+2}{2^n}\} \to -\frac{3}{2}, 1, -\frac{5}{8}, \frac{3}{8}, -\frac{7}{32}$

## Where points get lost

- Writing only $+\sqrt{}$ for a whole hyperbola (leave it in "$= 1$" form or keep $\pm$).
- Arithmetic slips while simplifying, like $\left(\frac{x+3}{-2}\right)^2$ with $-2$ on the bottom instead of 4: check one $t$-chart point in the final equation.
- Recursive: starting the formula at the wrong $n$.